

$$\begin{aligned}
& \frac{1}{16} g_2^4 y_u^{i12} \left(\frac{1}{16} g_2^4 y_u^{i12} L_{F3,0} [m_2] + \frac{1}{6} g_2^4 y_u^{i12} L_{F4,-1} [m_2] - \right. \\
& \frac{8}{45} g_2^4 y_u^{i12} L_{F5,-2} [m_2] - \frac{5}{72} \sum_p g_2^4 y_u^{i12} L_{F3,0} [m_p^i] + \frac{1}{48} \sum_p g_2^4 y_u^{i12} L_{F4,-1} [m_p^i] + \\
& \frac{2}{45} \sum_p g_2^4 y_u^{i12} L_{F5,-2} [m_p^i] + \frac{1}{24} g_2^2 y_u^{i12} (36 \overline{y_u^P} y_u^P - 5 \sum_p g_2^2) L_{F3,0} [m_p^i] + \\
& \frac{1}{16} g_2^2 y_u^{i12} (-24 \overline{y_u^P} y_u^P + \sum_p g_2^2) L_{F4,-1} [m_p^i] + \frac{1}{15} \sum_p g_2^4 y_u^{i12} L_{F5,-2} [m_p^i] + \\
& \frac{1}{18} g_2^4 y_u^{i12} L_{F3,0} [\tilde{\mu}] + \frac{1}{12} g_2^4 y_u^{i12} L_{F4,-1} [\tilde{\mu}] - \frac{4}{45} g_2^4 y_u^{i12} L_{F5,-2} [\tilde{\mu}] - \\
& \frac{1}{36} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_1, m_q^P] + \frac{1}{18} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,-1} [m_1, m_q^P] - \\
& \frac{1}{36} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F4,1,-2} [m_1, m_q^P] - \frac{1}{864} g_1^2 y_u^{i12} (g_1^2 - 3 g_2^2) L_{F2,2,-1} [m_1, m_q^{i1}] - \\
& \frac{4}{9} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_1, m_u^r] + \frac{8}{9} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,-1} [m_1, m_u^r] - \\
& \frac{4}{9} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F4,1,-2} [m_1, m_u^r] + \frac{2}{27} g_1^4 y_u^{i12} L_{F2,2,-1} [m_1, m_u^{i2}] - \\
& \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_2, m_q^P] + \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,-1} [m_2, m_q^P] - \\
& \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F4,1,-2} [m_2, m_q^P] - \frac{1}{32} g_2^2 y_u^{i12} (g_1^2 + g_2^2) L_{F2,2,-1} [m_2, m_q^{i1}] - \\
& \frac{4}{3} g_2^4 y_u^{i12} L_{F2,1,0} [m_2, \tilde{\mu}] - \frac{17}{12} g_2^4 y_u^{i12} L_{F2,2,-1} [m_2, \tilde{\mu}] + \frac{1}{3} g_2^4 y_u^{i12} L_{F3,1,-1} [m_2, \tilde{\mu}] + \\
& \frac{1}{3} g_2^4 y_u^{i12} L_{F3,2,-2} [m_2, \tilde{\mu}] - \frac{3}{2} g_2^4 m_2^2 y_u^{i12} L_{F3,2,-1} [m_2, \tilde{\mu}] - g_2^4 y_u^{i12} L_{F3,3,-3} [m_2, \tilde{\mu}] + \\
& g_2^4 m_2^2 y_u^{i12} L_{F3,3,-2} [m_2, \tilde{\mu}] - g_2^4 y_u^{i12} L_{F4,2,-3} [m_2, \tilde{\mu}] + g_2^4 m_2^2 y_u^{i12} L_{F4,2,-2} [m_2, \tilde{\mu}] - \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_3, m_q^P] + \frac{8}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,-1} [m_3, m_q^P] - \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F4,1,-2} [m_3, m_q^P] - \frac{1}{18} g_3^2 y_u^{i12} (g_1^2 - 3 g_2^2) L_{F2,2,-1} [m_3, m_q^{i1}] - \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_3, m_u^r] + \frac{8}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,-1} [m_3, m_u^r] - \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F4,1,-2} [m_3, m_u^r] + \frac{2}{9} g_1^2 g_3^2 y_u^{i12} L_{F2,2,-1} [m_3, m_u^{i2}] - \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} L_{F2,2,0} [m_d^r, m_q^P] + \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} L_{F2,1,0} [m_d^r, \tilde{\mu}] - \\
& \frac{1}{24} g_1^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} L_{F2,2,-1} [m_d^r, \tilde{\mu}] - \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_e^P} y_e^P y_u^{i12} L_{F2,2,0} [m_e^r, m_q^P] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_e^P} y_e^P y_u^{i12} L_{F3,1,0} [m_e^r, m_e^r] + \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_e^P} y_e^P y_u^{i12} L_{F3,2,-1} [m_e^r, m_e^r] - \\
& \frac{3}{8} \tilde{\mu}^2 \overline{y_e^P} y_e^P y_u^{i12} (3 g_1^2 + g_2^2) L_{F4,1,-1} [m_e^r, m_e^r] + \\
& \frac{1}{6} \tilde{\mu}^2 \overline{y_e^P} y_e^P y_u^{i12} (3 g_1^2 + g_2^2) L_{F5,1,-2} [m_e^r, m_e^r] + \\
& \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} L_{F3,1,0} [m_q^P, m_d^r] + \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} L_{F3,2,-1} [m_q^P, m_d^r] - \\
& \frac{3}{8} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} (5 g_1^2 + 3 g_2^2) L_{F4,1,-1} [m_q^P, m_d^r] + \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{i12} (3 g_1^2 + g_2^2) L_{F5,1,-2} [m_q^P, m_d^r] - \\
& \frac{3}{2} \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{sR} y_u^{i12} L_{F2,1,0} [m_q^P, m_q^S] + \frac{3}{2} \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{sR} y_u^{i12} L_{F3,1,-1} [m_q^P, m_q^S] - \\
& \frac{1}{8} y_u^{i12} \overline{a_u^P} a_u^P (g_1^2 - 7 g_2^2) L_{F2,2,0} [m_q^P, m_u^r] + \frac{3}{2} g_2^2 y_u^{i12} \overline{a_u^P} a_u^P L_{F3,1,0} [m_q^P, m_u^r] - \\
& g_2^2 y_u^{i12} \overline{a_u^P} a_u^P L_{F3,2,-1} [m_q^P, m_u^r] - \frac{3}{2} g_2^2 y_u^{i12} \overline{a_u^P} a_u^P L_{F4,1,-1} [m_q^P, m_u^r] + \\
& \frac{1}{12} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_q^P, \tilde{\mu}] - \frac{1}{24} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,2,-1} [m_q^P, \tilde{\mu}] + \\
& \frac{1}{432} g_1^2 y_u^{i12} (g_1^2 - 3 g_2^2) L_{F2,1,0} [m_q^{i1}, m_1] + \frac{1}{16} g_2^2 y_u^{i12} (g_1^2 + g_2^2) L_{F2,1,0} [m_q^{i1}, m_2] + \\
& \frac{1}{9} g_2^2 y_u^{i12} (g_1^2 - 3 g_2^2) L_{F2,1,0} [m_q^{i1}, m_3] + \frac{1}{4} y_u^{i12} \overline{a_u^P} a_u^P (7 g_1^2 + g_2^2) L_{F3,1,0} [m_u^r, m_q^P] + \\
& \frac{1}{8} y_u^{i12} \overline{a_u^P} a_u^P (g_1^2 - 7 g_2^2) L_{F3,2,-1} [m_u^r, m_q^P] - \frac{3}{4} y_u^{i12} \overline{a_u^P} a_u^P (5 g_1^2 + g_2^2) \\
& L_{F4,1,-1} [m_u^r, m_q^P] + \frac{1}{2} y_u^{i12} \overline{a_u^P} a_u^P (3 g_1^2 + g_2^2) L_{F5,1,-2} [m_u^r, m_q^P] - \\
& \frac{1}{6} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,0} [m_u^r, \tilde{\mu}] + \frac{1}{12} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,2,-1} [m_u^r, \tilde{\mu}] - \\
& \frac{4}{27} g_1^4 y_u^{i12} L_{F2,1,0} [m_u^{i2}, m_1] - \frac{4}{9} g_1^2 g_2^2 y_u^{i12} L_{F2,1,0} [m_u^{i2}, m_3] - \\
& \frac{1}{2} g_1^2 y_u^{i12} (g_1^2 + g_2^2) L_{F3,1,-1} [\tilde{\mu}, m_1] + \frac{3}{8} g_1^4 y_u^{i12} L_{F3,2,-2} [\tilde{\mu}, m_1] + \\
& \frac{3}{8} g_1^4 m_1^2 y_u^{i12} L_{F3,2,-1} [\tilde{\mu}, m_1] + \frac{1}{12} g_1^2 y_u^{i12} (12 g_1^2 + 7 g_2^2) L_{F4,1,-2} [\tilde{\mu}, m_1] - \\
& \frac{1}{4} g_1^4 y_u^{i12} L_{F4,2,-3} [\tilde{\mu}, m_1] - \frac{1}{4} g_1^4 m_1^2 y_u^{i12} L_{F4,2,-2} [\tilde{\mu}, m_1] - \\
& \frac{1}{6} g_1^2 y_u^{i12} (3 g_1^2 + g_2^2) L_{F5,1,-3} [\tilde{\mu}, m_1] - \frac{1}{6} g_1^2 y_u^{i12} (9 g_1^2 + g_2^2) L_{F3,1,-1} [\tilde{\mu}, m_2] + \\
& \frac{85}{24} g_2^4 y_u^{i12} L_{F3,2,-2} [\tilde{\mu}, m_2] + \frac{3}{8} g_2^4 m_2^2 y_u^{i12} L_{F3,2,-1} [\tilde{\mu}, m_2] + \\
& \frac{3}{2} g_2^2 y_u^{i12} (4 g_1^2 + g_2^2) L_{F4,1,-2} [\tilde{\mu}, m_2] - \frac{7}{4} g_2^4 y_u^{i12} L_{F4,2,-3} [\tilde{\mu}, m_2] - \\
& \frac{1}{4} g_2^4 m_2^2 y_u^{i12} L_{F4,2,-2} [\tilde{\mu}, m_2] - \frac{1}{2} g_2^2 y_u^{i12} (3 g_1^2 + g_2^2) L_{F5,1,-3} [\tilde{\mu}, m_2] - \\
& \frac{1}{2} \overline{y_d^P} y_d^S \overline{y_u^S} y_u^{p12} y_u^{i1t} L_{F2,1,0} [\tilde{\mu}, m_d^r] - \overline{y_d^P} y_d^S \overline{y_u^S} y_u^{p12} y_u^{i1t} L_{F3,1,-1} [\tilde{\mu}, m_d^r] - \\
& \frac{1}{2} \overline{y_d^P} y_d^S \overline{y_u^S} y_u^{p12} y_u^{i1t} L_{F4,1,-2} [\tilde{\mu}, m_d^r] - \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F2,1,0} [\tilde{\mu}, m_q^P] + \\
& 2 \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F3,1,-1} [\tilde{\mu}, m_q^P] + \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F4,1,-2} [\tilde{\mu}, m_q^P] - \\
& \frac{1}{2} \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F2,1,0} [\tilde{\mu}, m_u^t] + \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F3,1,-1} [\tilde{\mu}, m_u^t] - \\
& \frac{1}{2} \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F4,1,-2} [\tilde{\mu}, m_u^t] + \frac{1}{9} m_1 g_1^2 y_u^{p12} y_u^{i1r} \overline{a_u^P} L_{F2,1,1,0} [m_1, m_q^P, m_u^r] - \\
& \frac{1}{9} m_1 g_1^2 y_u^{p12} y_u^{i1r} \overline{a_u^P} L_{F3,1,1,-1} [m_1, m_q^P, m_u^r] + \frac{1}{18} m_1 g_1^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} \\
& L_{F2,1,1,0} [m_1, m_q^P, m_u^{i2}] + \frac{1}{36} m_1 g_1^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} L_{F2,2,1,-1} [m_1, m_q^P, m_u^{i2}] - \\
& \frac{1}{18} m_1 g_1^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} L_{F3,1,1,-1} [m_1, m_q^P, m_u^{i2}] + \frac{1}{6} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_1, m_q^P, \tilde{\mu}] - \\
& \frac{1}{3} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_1, m_q^P, \tilde{\mu}] + \frac{1}{6} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,1,-2} [m_1, m_q^P, \tilde{\mu}] + \\
& \frac{1}{36} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_1, m_q^{i1}, m_q^P] - \frac{1}{72} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} \\
& L_{F2,1,1,-1} [m_1, m_q^{i1}, m_q^P] + \frac{1}{18} m_1 g_1^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,1,1,0} [m_1, m_q^{i1}, m_u^r] - \\
& \frac{1}{36} m_1 g_1^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,2,1,-1} [m_1, m_q^{i1}, m_u^r] - \frac{1}{18} m_1 g_1^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} \\
& L_{F3,1,1,-1} [m_1, m_q^{i1}, m_u^r] - \frac{1}{72} m_1 g_1^2 a_u^{i12} (g_1^2 + g_2^2) L_{F2,2,1,-1} [m_1, m_q^{i1}, m_u^{i2}] + \\
& \frac{1}{48} g_1^4 y_u^{i12} L_{F2,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] + \frac{1}{72} g_1^4 m_1^2 y_u^{i12} L_{F2,1,1,0} [m_1, m_q^{i1}, \tilde{\mu}] - \\
& \frac{1}{72} g_1^4 y_u^{i12} L_{F3,1,1,-2} [m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{72} g_1^4 m_1^2 y_u^{i12} L_{F3,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{36} m_1 g_1^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,2,1,-1} [m_1, m_u^r, m_q^{i1}] - \frac{2}{3} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_1, m_u^r, \tilde{\mu}] + \\
& \frac{4}{3} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_1, m_u^r, \tilde{\mu}] - \frac{2}{3} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,1,-2} [m_1, m_u^r, \tilde{\mu}] - \\
& \frac{1}{36} m_1 g_1^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} L_{F2,2,1,-1} [m_1, m_u^{i2}, m_q^P] - \\
& \frac{1}{72} m_1 g_1^2 a_u^{i12} (g_1^2 + g_2^2) L_{F2,2,1,-1} [m_1, m_u^{i2}, m_q^{i1}] + \\
& \frac{4}{9} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_1, m_u^{i2}, m_u^r] - \frac{2}{3} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_1, m_u^{i2}, m_u^r] + \\
& \frac{1}{3} g_1^4 y_u^{i12} L_{F2,1,1,-1} [m_1, m_u^{i2}, \tilde{\mu}] + \frac{2}{9} g_1^4 m_1^2 y_u^{i12} L_{F2,1,1,0} [m_1, m_u^{i2}, \tilde{\mu}] - \\
& \frac{2}{9} g_1^4 y_u^{i12} L_{F3,1,1,-2} [m_1, m_u^{i2}, \tilde{\mu}] - \frac{2}{9} g_1^4 m_1^2 y_u^{i12} L_{F3,1,1,-1} [m_1, m_u^{i2}, \tilde{\mu}] + \\
& \frac{1}{144} g_1^2 (6 \overline{y_u^P} y_u^{p12} y_u^{i1r} + y_u^{i12} (8 g_1^2 - 3 g_2^2)) L_{F2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{11}{144} g_1^4 m_1^2 y_u^{i12} L_{F2,2,1,-1} [m_1, \tilde{\mu}, m_q^{i1}] - \frac{5}{8} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,2,1,-2} [m_1, \tilde{\mu}, m_u^r] + \\
& \frac{1}{36} g_1^2 (-6 \overline{y_u^P} y_u^{p12} y_u^{i1r} + y_u^{i12} (-13 g_1^2 + 3 g_2^2)) L_{F2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i2}] - \\
& \frac{4}{9} g_1^4 m_1^2 y_u^{i12} L_{F2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i2}] - \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_2, m_q^P, \tilde{\mu}] + \\
& 3 g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_2, m_q^P, \tilde{\mu}] - \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F3,1,1,-2} [m_2, m_q^P, \tilde{\mu}] + \\
& \frac{1}{4} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_2, m_q^{i1}, m_q^P] - \frac{1}{8} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_2, m_q^{i1}, m_q^P] + \\
& \frac{15}{16} g_2^4 y_u^{i12} L_{F2,1,1,-1} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 y_u^{i12} L_{F2,1,1,0} [m_2, m_q^{i1}, \tilde{\mu}] - \\
& \frac{5}{8} g_2^4 y_u^{i12} L_{F3,1,1,-2} [m_2, m_q^{i1}, \tilde{\mu}] - \frac{1}{8} g_2^4 m_2^2 y_u^{i12} L_{F2,1,1,-1} [m_2, m_q^{i1}, \tilde{\mu}] + \\
& 2 g_1^2 g_2^2 y_u^{i12} L_{F2,2,1,-2} [m_2, \tilde{\mu}, m_1] - \frac{1}{2} m_2 m_1 g_1^2 g_2^2 y_u^{i12} L_{F2,2,1,-1} [m_2, \tilde{\mu}, m_1] - \\
& g_1^2 g_2^2 y_u^{i12} L_{F3,2,1,-3} [m_2, \tilde{\mu}, m_1] + \frac{1}{2} m_2 m_1 g_1^2 g_2^2 y_u^{i12} L_{F3,2,1,-2} [m_2, \tilde{\mu}, m_1] - \\
& \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{p12} L_{F2,2,1,-1} [m_2, \tilde{\mu}, m_q^P] - \frac{3}{2} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,2,1,-2} [m_2, \tilde{\mu}, m_q^P] + \\
& \frac{1}{16} g_2^2 (-6 \overline{y_u^P} y_u^{p12} y_u^{i1r} + y_u^{i12} (3 g_1^2 - 22 g_2^2)) L_{F2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] - \\
& \frac{5}{16} g_2^4 m_2^2 y_u^{i12} L_{F2,2,1,-1} [m_2, \tilde{\mu}, m_q^{i1}] - \frac{1}{8} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,2,1,-2} [m_2, \tilde{\mu}, m_u^r] + \\
& \frac{4}{3} m_3 g_3^2 y_u^{p12} y_u^{i1r} \overline{a_u^P} L_{F2,1,1,0} [m_3, m_q^P, m_u^r] - \\
& \frac{4}{3} m_3 g_3^2 y_u^{p12} y_u^{i1r} \overline{a_u^P} L_{F3,1,1,-1} [m_3, m_q^P, m_u^r] + \\
& \frac{2}{3} m_3 g_3^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} L_{F2,1,1,0} [m_3, m_q^P, m_u^{i2}] + \frac{1}{3} m_3 g_3^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} \\
& L_{F2,2,1,-1} [m_3, m_q^P, m_u^{i2}] - \frac{2}{3} m_3 g_3^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} L_{F3,1,1,-1} [m_3, m_q^P, m_u^{i2}] + \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_3, m_q^{i1}, m_q^P] - \frac{2}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_3, m_q^{i1}, m_q^P] + \\
& \frac{2}{3} m_3 g_3^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,1,1,0} [m_3, m_q^{i1}, m_u^r] - \\
& \frac{1}{3} m_3 g_3^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,2,1,-1} [m_3, m_q^{i1}, m_u^r] - \frac{2}{3} m_3 g_3^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} \\
& L_{F3,1,1,-1} [m_3, m_q^{i1}, m_u^r] - \frac{1}{6} m_3 g_3^2 a_u^{i12} (g_1^2 + g_2^2) L_{F2,2,1,-1} [m_3, m_q^{i1}, m_u^{i2}] + \\
& \frac{1}{3} m_3 g_3^2 \overline{y_u^P} y_u^{p12} a_u^{i1r} L_{F2,2,1,-1} [m_3, m_u^r, m_q^{i1}] - \frac{1}{3} m_3 g_3^2 \overline{y_u^P} y_u^{i1r} a_u^{p12} \\
& L_{F2,2,1,-1} [m_3, m_u^{i2}, m_q^P] - \frac{1}{6} m_3 g_3^2 a_u^{i12} (g_1^2 + g_2^2) L_{F2,2,1,-1} [m_3, m_u^{i2}, m_q^{i1}] + \\
& \frac{4}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F1,1,1,0} [m_3, m_u^{i2}, m_u^r] - \frac{2}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_3, m_u^{i2}, m_u^r] - \\
& \frac{1}{24} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{p12} (g_1^2 - 3 g_2^2) L_{F2,1,1,0} [m_d^r, m_q^P, \tilde{\mu}] - \\
& \frac{1}{6} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{p12} (g_1^2 + g_2^2) L_{F3,1,1,-1} [m_d^r, m_q^P, \tilde{\mu}] - \\
& \frac{1}{16} \tilde{\mu}^2 \overline{y_d^P} (y_d^{i1r} (y_u^{p12} (g_1^2 + g_2^2) + 2 \overline{y_u^S} y_u^{pT} y_u^{s12}) - 2 y_d^S \overline{y_u^S} y_u^{p12} y_u^{i1t}) \\
& L_{F2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^P] + \frac{1}{72} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^P, m_1, m_q^{i1}] + \\
& \frac{1}{6} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^P, m_2, m_q^{i1}] + \frac{2}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^P, m_3, m_q^{i1}] + \\
& \frac{1}{24} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{p12} (g_1^2 - 3 g_2^2) L_{F2,1,1,0} [m_q^P, m_d^r, \tilde{\mu}] - \\
& \frac{1}{6} \tilde{\mu}^2 \overline{y_d^P} y_d^P y_u^{p12} (g_1^2 + g_2^2) L_{F3,1,1,-1} [m_q^P, m_d^r, \tilde{\mu}] + \\
& \frac{3}{2} \tilde{\mu}^2 \overline{y_d^P} y_d^S \overline{y_u^S} y_u^{pT} y_u^{i12} L_{F2,2,1,-1} [m_q^P, m_q^S, m_d^r] - \\
& \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S L_{F2,1,1,0} [m_q^P, m_q^S, m_u^r] + \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S \\
& L_{F3,1,1,-1} [m_q^P, m_q^S, m_u^r] + \frac{1}{2} \overline{y_u^P} \overline{y_u^S} y_u^{pT} y_u^{s12} y_u^{i1r} L_{F1,1,1,0} [m_q^P, m_q^S, \tilde{\mu}] + \\
& \frac{1}{4} \overline{y_u^P} \overline{y_u^S} y_u^{pT} (y_u^{p12} y_u^{s12} y_u^{i1r} - y_u^{p12} y_u^{i1r} y_u^{i1t}) L_{F2,1,1,-1} [m_q^P, m_q^S, \tilde{\mu}] - \\
& \frac{1}{16} \tilde{\mu}^2 \overline{y_d^P} (y_u^{i1r} (y_u^{p12} (g_1^2 + g_2^2) - 2 \overline{y_u^S} y_u^{pT} y_u^{s12}) + 2 y_d^S \overline{y_u^S} y_u^{p12} y_u^{i1t}) \\
& L_{F2,2,1,-1} [m_q^P, \tilde{\mu}, m_d^r] - \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S L_{F2,1,1,0} [m_q^S, m_q^P, m_u^r] + \\
& \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S L_{F3,1,1,-1} [m_q^S, m_q^P, m_u^r] + \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S \\
& L_{F2,2,1,-1} [m_q^S, m_u^r, m_q^P] - \frac{1}{72} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^{i1}, m_1, m_q^P] + \\
& \frac{1}{108} m_1 g_1^2 a_u^{i12} (g_1^2 + 9 g_2^2) L_{F2,1,1,0} [m_q^{i1}, m_1, m_u^{i2}] - \frac{1}{36} m_1 g_1^2 a_u^{i12} (g_1^2 + g_2^2) \\
& L_{F3,1,1,-1} [m_q^{i1}, m_1, m_u^{i2}] + \frac{1}{72} g_1^2 y_u^{i12} (g_1^2 - 3 g_2^2) L_{F2,1,1,-1} [m_q^{i1}, m_1, \tilde{\mu}] - \\
& \frac{1}{8} g_2^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^{i1}, m_2, m_q^P] - \frac{1}{8} g_2^2 y_u^{i12} (g_1^2 + g_2^2) L_{F2,1,1,-1} [m_q^{i1}, m_2, \tilde{\mu}] - \\
& \frac{2}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_q^{i1}, m_3, m_q^P] + \frac{1}{9} m_3 g_3^2 a_u^{i12} (g_1^2 + 9 g_2^2) \\
& L_{F2,1,1,0} [m_q^{i1}, m_3, m_u^{i2}] - \frac{1}{3} m_3 g_3^2 a_u^{i12} (g_1^2 + g_2^2) L_{F3,1,1,-1} [m_q^{i1}, m_3, m_u^{i2}] + \\
& \frac{2}{9} g_1^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_u^r, m_1, m_u^{i2}] + \frac{2}{3} g_3^2 \overline{y_u^P} y_u^{p12} y_u^{i1r} L_{F2,1,1,-1} [m_u^r, m_3, m_u^{i2}] - \\
& \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S L_{F2,1,1,0} [m_u^r, m_q^P, m_q^S] + \\
& \frac{3}{2} \overline{y_u^S} y_u^{pT} y_u^{i12} \overline{a_u^P} a_u^S L_{F3,1,1,-1} [m_u^r, m_q^P, m_q^S] - \\
& 3 \overline{y_u^S} y_u^S y_u^{i12} \overline{a_u^P} a_u^P L_{F2,1,1,0} [m_u^r, m_q^P, m_u^t] + \\
& 3 \overline{y_u^S} y_u^S y_u^{i12} \overline{a_u^P} a_u^P L_{F3,1,1,-1} [m_u^r, m_q^P, m_u^t] + \\
& \frac{3}{2} \overline{y_u^S} y_u^S y_u^{i12} \overline{a_u^P} a_u^P L_{F2,2,1,-1} [m_u^r, m_u^t, m_q^P] + \\
& \frac{1}{2} \overline{y_u^P} \overline{y_u^S} y_u^{p12} y_u^{sR} y_u^{i1t} L_{F1,1,1,0} [m_u^r, m_u^t, \tilde{\mu}] + \\
& \frac{1}{4} \overline{y_u^P} \overline{y_u^S} y_u^{p12} y_u^{sR} y_u^{i1t} L_{F2,1,1,-1} [m_u^r, m_u^t, \tilde{\mu}] - \\
& \frac{1}{4} \overline{y_u^P} \overline{y_u^S} y_u^{p12} y$$